ZOO-504 Addition 2 (16 jan 2022)

Current Paper ADMIRE group of Zoologist

Sir Arslan

1. National park define.

National Park

According to the 'Modern Protected Areas' legislation, a national park is a protected area set aside by the government for the protection and conservation of its outstanding scenery and wildlife in a natural state. It is accessible to public for research, education and recreation.

Some of the names of Pakistan national park given as under

1. Ayub Park Rawalpindi
2. Shandure-Phander National Park Ghizer
3. Broghil Valley Chitral
4. Central Karakoram Gilgit and Skardu
5. Chinji Chakwal

2. Hunting of markhor.

Markhor is a national animal of Pakistan. Markhor hunting in Pakistan's Gilgit-Baltistan region has become a lucrative business for the locals. A Spanish citizen became the third foreign hunter of the season by hunting a Markhor – the national animal of Pakistan, on the outskirts of Gilgit-Baltistan.

3. What is icons

The definition of an icon is a graphic representation of something, a person or thing that is symbolic or is a noted figure. An example of an icon is the "home" or "finder" icon on your computer.

4. Define territory ?

A geographic area belonging to or under the jurisdiction of a governmental authority

5. Economic important of Wild Life? (5)

From the economic point of view wildlife is important and serves as source of

• **Tourism**

Fascination with the beauty of wild animals drives tourism worldwide. e.g. Safari park in Lahore Pakistan. Kruger National Park in South Africa.

• **Animal Protein**

e.g. In some forested area in Latin America, wildlife acts as animal protein source.

• **Bushmeat** (Wild meat)

Wild meat tends to have less fat than domestic meat and can be a good source of iron, Vitamin A and Vitamin B.

• **Trading of Bushmeat**

In sub-Saharan Africa there is a long tradition of trade between rural areas and the major towns, where it is sold as a high-priced delicacy.

• **Ranching and Farming**

Commercial ranching and farming of wild animal occurring in countries like China, Zimbabwe & Thailand.

• **Traditional medicine/ Pharmaceutics**

e.g. *Nerium oleander* (Anvirzel)

• **Wood**

Wildlife resources like wood from trees are used for making houses.

• Silk
• Paper
• Leather/Hide
• Ivory

6. Run of Kutch?

The Great Rann of Kutch is a salt marsh. Located in the Thar Desert. Kutch District of Gujarat, India and the Sindh, Province of Pakistan. Reputed to be one of the largest salt deserts in the world.

7. Margalla hills.

Location:

North of the Islamabad City, Capital of Pakistan.

Area

• 12,000 ha

Features:

• Elevation range 550m to 1500m.
• Rugged topography
• Numerous valleys
• Precipitous slopes
• Margalla hills have beautiful torrents gushing down in the monsoon.
• Natural springs are also present.
• The Park includes the Margalla Hills which form's the foothills of the Himalayas, along with Shakarparian Park and Rawal Lake.

8. Mintaka

The name of the game reserve is actually driven from *"The Mintaka Pass or Mingteke Pass or Mintika Pass" and the nearby "Kilik Pass"*

Mintaka means

" Thousand goats.

Location

Kilik/Mintaka Game Reserve lies along the border with China. East of the Korakorum highway and the Khunjerab National Park.

Geographical Location:

Lies in Hunza, Gilgit District, on the Pakistan-China boader, 225km from the town of Gilgilt.

Area:

It includes 65,036ha

Features

• The terrain is rugged and mountainous
• There is a variety of rocks o Sedimentary o Metamorphic o Igneous

Mixture of rock and mineral deposits

9. Indus River Dolphin Conservation Project (IRDCP). (5)

Introduction:

Indus River Dolphin *(Platanista gangetica minor)*, an endangered species endemic to the Indus River system in Pakistan. This project was developed to conserve a viable population

Habitat: Habitat has been reduced to one fifth of its historical range and further degraded due to shortage of water.

Population: Divided in sub-populations because of the six barrages constructed on the River Indus.

Project Area: Covering an area of 200 km between Guddu and Sukkur barrages in the province of Sindh and some extensions of the project covers the Punjab.

Project Objectives:

• Protecting the innate biodiversity of the Lower Indus River Basin Eco-system.
• Ensuring the sustainable use of riverine biodiversity.
• Promoting actions to mitigate pollution and the wasteful extraction of riverine resources.

Project Sponsors:

• WWF-Switzerland
• Swedish International Development Agency (SIDA)
• Engro-foods Limited, Pakistan.

Stakeholders:

WWF-Pakistan
• Sindh Wildlife Department (SWD)
• Agriculture Extension Department, Sindh (AED)
• Sindh Environmental Protection Agency (SEPA)
• Sindh Fisheries Department (SFD)

Project Approach:

The Indus River Dolphin Conservation Project focuses on the root causes of biodiversity loss by linking the protection of the Indus River Dolphin with measures in the agricultural and fisheries sectors.

10. Exotic species?

exotic species refers to a plant species or an animal species that is non-native. It is introduced into an area where it does not occur naturally.

11. Chasma tonsa barrage Dolphin sanctuary.

Chashma and Taunsa Barrage Dolphin Sanctuary was declared open to the public in 1972

Location:

• Mianwali District, Punjab-Pakistan

Features:

For a wide variety of waterfowls including threatened species, an important site for

Breeding
✦ Staging
✦ Wintering
✦ Mid-winter waterfowl counts in recent years have regularly exceeded 20,000 birds
✦ Anatidae (ducks, geese, swans)
✦ The endangered Indus Dolphin occurs in the river both upstream and downstream of the barrage

Issues for consideration:

• Taunsa Barrage Wildlife Sanctuary is currently confronted with multiple issues
• Limited cultivation is practiced inside the sanctuary area
• Clearing off the natural vegetation

• Alteration of the natural ecosystem
• Degrading wildlife habitat
• Invasion of alien species
• Hence, adverse environmental impacts.

Management Plan:

WWF-Pakistan has developed a sustainable management plan. Indus river dolphin conservation project.

12.Importance of hawk and eagle?

Eagles, hawks and their feathers have great cultural and spiritual value to Native Americans as religious objects.

13.Musk deer? (L-5)

IUCN Red list Threat Status: Endangered

Musk deer can refer to any one, or all seven, of the species that make up Moschus, the only extant genus of the family Moschidae.

The musk deer are lacking antlers.

Description

Markedly different from other species of deer. No glands on face and foot. Females have only two mammary glands. 50 cm high and weighs about 11-12 kg. Hind legs are much longer than the fore legs. Ears very long and prominent. Tail is very short and completely hidden in anal hair. Body covered with thick long bristly hair. Color greyish brown with grey spots on the back. Male has downwardly projecting canines about 8 cm in length and are used in fighting. Musk is present in the musk pouch which is present below the skin of the abdomen of male only.

Distribution

In Pakistan it is found in sub alpine zones of Himalayas. At about 2500-4000 m altitude in Chitral, Gilgit, Hazara and Azad Kashmir.

Threats

There is a high trade in Musk deer parts, particularly pods, into China and elsewhere in northeast Asia. Hunted for meat, which is considered a delicacy locally. Trade of musk glands, which has led to a substantial increase in hunting since long. The musk is valued for its cosmetic and pharmaceutical properties. Yield of the musk is only about 25 grams

Conservation actins

This species is listed on CITES Appendix I. The high value of the parts in trade mean that conservation requires effective hand-on anti-poaching activity.

14.Ramsar sites in Punjab?

There are three ramsar sites in Pakistan

Punjab

• Uchhali Complex
• Taunsa Barrage
• Chashma Barrage

15.Role of insects and birds in maintenance of ecosystem? (5)

Role of Insects :

Insects **are** crucial components of many ecosystems, **where** they perform many important **functions**. They aerate the soil, pollinate blossoms, and control insect and plant **pests**.

Many insects**,** especially beetles, are scavengers, feeding on dead animals and fallen trees, thereby recycling nutrients back into the soil.

Insects are **important** for pollinating crops we use for food, as a food source themselves in some countries, and decomposing dead material to recycle nutrients. **Insects** are also **important** because of the damage they can cause to humans.

Role of Birds

In many cases the activities of birds can have large consequences for the ecosystems they inhabit, making them incredibly important in the overall functioning of various ecosystems.

By contributing in such an important way to ecosystem health, birds can provide a number of direct benefits to humans. The Millennium Ecosystem Assessment, a study initiated by the United Nations, coined the term “ecosystem services” to describe these kinds of services.

16.Positive values of wildlife?

People find many reasons to value wildlife. Virtually everyone appreciates the aesthetic value of natural beauty or artistic appeal present in animal life. Giant pandas, bald eagles, and infant harp seals are familiar examples of wildlife with outstanding aesthetic value. Wild species offer recreational value, the most common examples of which are sport hunting and bird watching. Wildlife also has utilization value which results from its practical uses . Examples of utilitarian value range from genetic reservoirs for crop and livestock improvement to diverse biomedical and pharmaceutical users . A related category commercial value , includes such familiar examples as the sale of furs and hunting leases.

17.Tooshi game reserve 2

Shandur

• Broghil National Park
• Mastuj
• Booni

- Bamborat: area where the Kalash cultural festival is held every August of the year.

Location Tooshi Game Reserve, is located along the paved road to Garam Chashma in Chitral's Lutkho district.

Area:

It covers 1000 hectares.

Features :

It is proposed to be reclassified as a wildlife sanctuary, with readily viewable Markhor population. The jeep ride is just 20 minutes from Chitral town. The Markhor are best seen along the road at dusk. In the late afternoon until dusk, Markhor can be seen while descending to drink water from the passing river

18.Cavity trees 3

Cavity Trees:

• Holes excavated in snags by woodpeckers. Shelter and nesting cover by many species.

19.Protected area with use of sustainable resources ? 5

Protected Area with Sustainable Use of Natural Resources Definition:

Protected areas that conserve ecosystems and habitats, together with associated cultural values and traditional natural resource management systems. Generally large, with most of the area in a natural condition. Where a proportion is under sustainable natural resource management. Where low-level non-industrial use of natural resources compatible with nature conservation is seen as one of the main aims of the area.

Primary objective

• To protect natural ecosystems

• Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

Other objectives

• To promote sustainable use of natural resources.

• To promote social and economic benefits to local communities.

• To facilitate scientific research and environmental monitoring.

• To facilitate recreation and appropriate small-scale tourism.

Distinguishing features

• It is not designed to accommodate large-scale industrial harvest.

• Achieve nature conservation.

• Category VI protected areas aim to conserve ecosystems and habitats

Role in the landscape/seascape

• Protected areas are particularly adapted to the application of landscape approaches.

• It is particularly appropriate to the conservation of natural ecosystems when there are few or no areas without use or occupation

20.Hub dam wildlife sanctuary 3

Hab Dam Wildlife Sanctuary was declared in July 1974. It is Pakistan's third largest dam

Area:

27,219ha

Location:

It is situated in the north of Karachi bordering the Kirthar National Park Balochistan

Features:

• Hab Dam Wildlife Sanctuary is covering an area of 32 square miles
• Much of the shoreline is steeply shelving and stony
• Many shallow bays
• Small islands
• The greater part of the reservoir (in Balochistan) is unprotected
• The Eastern shore and of the Dam (in Sindh) is protected in the Kirthar National

Fauna:

An ideal place for bird watching

Waterfowl (both resident and migratory). The lake also provide refuge to migratory birds

✦ Cranes
✦ Pelicans
✦ Ducks
✦ Waders

Fauna:

The surrounding hills are home for

- Urial
- Sindh Wild Goat
- Chinkara Gazelle
- Pangolin
- Wolf
- Jackal
- Common fox
- Numerous reptiles

21.How many sanctuaries in Pakistan? 2

In Pakistan presently there are 99 Wildlife Sanctuaries. IUCN recognized number of Wildlife Sanctuaries in Pakistan.

22.Ayubia national park ? 5

Ayubia National Park was declared a National Park in 1984. Ayubia was named after Muhammad Ayub Khan, the then President of Pakistan.

Location:

Abbottabad District, Khyber Pakhtunkhwa- Pakistan.

Area:

Initial area was 1684 ha, expanded through northern extension in 1998. Total area 3,312
hectares (8,184 acres)

Features:

• This Natiopnal park is surrounded by three small towns(Nathia gali, Ayubia and Khanaspur)
• Rainfall: 1,244 cm
• Temperature: 3 °C - 11 °C

The Area Supports

• Temperate coniferous forest
• Temperate broadleaf and mixed forest
• These forests have an average elevation of 8,000 feet (2,400 m) above the sea level.

Major Flora:

The park holds 104 species of plants. The main floral species are

• *Cedrus deodara*
• Blue pine
• Yew
• Silver fir
• Horse chestnut
• Oak

Plants belonging to 19 families are known for their medicinal properties.

Major Birds Fauna:

• The park harbors up to 203 species of birds.
• Golden eagle
• Vulture

• Eurasian sparrow
• Hawkt
• Collared dove

Major Mammal Fauna:

Ayubia National Park supports 31 species of mammals.

• Asiatic leopard
• Hill fox
• flying squirrel
• Masked civet
• Rhesus Macaque

23.Negative impacts of predation?

Negative impact of wildlife occurs when growing human populations overlap with established wildlife territory

. People lose their crops, livestock, property, and sometimes their lives
• Baboons in South Africa attacking young cattle.
• One-horned rhinos in Nepal destroying crops.
• Bears and wolves killing livestock.
• Wolf attacks sheep, and reindeers in Sweden.
The impacts of predation are huge. In Russian Far East farmers raise captive deer for human consumption. Deer are the natural prey preference for leopards, and in absence of wild prey, leopards venture into the deer farms in search of food. sources of pouutant.

24.Habitat/Species Management Area ? 5

Definition:
Protected areas aiming to protect particular species or habitats and management reflects this priority.
Primary objective
To maintain, conserve and restore species and habitats.
Other objectives
• To protect vegetation patterns.
• To protect fragments of habitats as components of landscape or seascapescale conservation strategies.
• To develop public education and appreciation of the species or habitat
Distinguishing features
Protection of particular species
Protection of habitats
• Active management to maintain target species
• Active management of culturally-defined ecosystems

Role in the landscape/seascape

• Protect critically endangered populations of species.
• Protect rare or threatened habitats.
• Maintain species that have become dependent on cultural landscapes where their original habitats have disappeared or been altered.

25.Khunjerab National Park note ?

Khunjerab National Park was established on 29 April 1979.

Location:

• Gilgit Baltistan-Pakistan.
• Adjacent to the Taxkorgan Natural Reserve in China.

Area:

Spreads over 2,26,913 ha

Features:

• One of the highest National Parks in the world.
• It is Pakistan's third largest National Park. Area of Park lies at an elevation of 4000m

Important Fact:

• The borders of the Park mapped in 1974, after a field survey to protect “Marcopolo Sheep”.
• The population of that sheep was around 400 but had dropped to below 180 by the time of the completion of the Karakoram Highway.

Major Fauna

The primary purpose of this park was to provide protection to the endangered **Marco Polo sheep**, which is only found in this area in Pakistan.

Snow leopards

• Worldwide---7,000-10,000
• Khunjerab Park---More than 300

Other Mammalian Fauna

• Himalayan ibex
• Himalayan brown bear
• Red fox
• Tibetan wolf
• Blue sheep
• Western kiang (Wild ass)
• Mountain weasel
• Beech marten
• Long-tailed marmot

Large-eared pika

• Cape hare
• Wood mouse
• Royle's mountain vole
• Asian house shrew
• Etruscan shrew
• Grey dwarf hamster

Birds Fauna

- Golden Eagle
- Black vulture
- Marsh Harrier
- Eurasian Sparrow Hawk
- Himalyan Snow Cock
- Snow Partridge
- Grey Heron
- Snow Pigeon

Important Fact:

According to WWF officials there is plan to turn “Khunjreb National Park” into “International Peace Park”. To conserve area on both sides of the boarder.

26. State causes of Wildlife depletion ?

Conflict between wildlife and human occur over

- Living space
- Food

These factors are the main threats to the continued survival of many species in different parts of the world.

The main factors of Persecution are:

Wildlife Trade

Trading of wildlife and its products. e.g. Bush meat, Ivory, hide, trophy, fur etc.

Trapping of Wild Animals

- Pets(cats etc.)
- Sold for Medical experimental studies (Monkeys, Rabbits etc.)
- For street shows, circuses (Bears,Parrots,Lions)

Poaching

- Illegal killing for tusks, oil, decoration (as stuffed preparation), Plumage.

Illegal hunting for bush meat, trophies.

Absurd killing

- Tigers, jackals, wolves etc. are killed by the farmers

27.Small Grants Programme ?

Small Grants Programme (SGP)

- *“Small support for long term impact”.*

SGP of WWF-Pakistan has been supporting grass root level organizations and researchers to strengthen efforts to conserve threatened and endangered species and ecosystem. Under the programme more than 450 projects have been successfully completed.

28. Primary objective of Astor Sanctuary?

Primary objective

The sanctuary is home to a small population of the near threatened species, Markhor. Markhor is National animal of Pakistan. It is a large species of wild goats unique to this region of the world.

29.Explain baltistan sanctuary (5)

Baltistan Wildlife Sanctuary was established in 1975

Location:

Skardu District. Between the villages of Rondu and Shengus, in the Skardu District. Baltistan region of Northern Pakistan South of the Indus River

Area:

41,500 ha

Important Fact

Baltistan Wildlife Sanctuary is contiguous with the Astore Wildlife Sanctuary to its South and East

Objective:

Purpose of conserving the threatened species

- Snow leopard
- Brown bear
- Lynx
- Tibetan wolf
- Tibetan sand fox
- Markhor
- Bharal
- Siberian ibex.

Avifauna:

Studies indicate that the area is rich in avifauna. 109 birds species have been recorded from the Deosai plateau. This a breeding ground for variety of animals

- Warblers
- Buntings
- Red start

30.Define scavengers (2)

Scavengers (Vultures, eagles, jackals, hyaenas etc.) & decomposers serves to perform important tasks.

31.What activities which impair the wetland vegetation (3)

Activities that can impair wetland vegetation

- Grazing by domestic animals.
- Introduction of nonnative plants that compete with natives.
- Removal of vegetation for peat mining

32. IUCN stand for? 2

International union of conservation of nature

33.Bushmeat ? 3

Bushmeat (Wild meat)

Wild meat tends to have less fat than domestic meat and can be a good source of iron, Vitamin A and Vitamin B.

- **Trading of Bushmeat**

In sub-Saharan Africa there is a long tradition of trade between rural areas and the major towns, where it is sold as a high-priced delicacy 20 mcqs

34. 5 ramsar sites of sindh? 5 marks

Sindh

- Keenjhar (Kalri) Lake
- Haleji Lake
- Drigh Lake
- Indus Dolphin reserve
- Jubho lagoon
- Nurri Lagoon
- Deh Akro-II
- Indus Delta
- Runn of Kutch

35.Criteria of wetland ? 5

Wetlands should be selected for the Ramsar List on account of their international significance in terms of the Biodiversity

Uniqueness in

- Ecology
- Flora
- Fauna

• Limnology or hydrology

There are nine criteria for identifying Wetlands of International Importance.

Group A

• Sites containing representative, rare or unique wetland types

Criterion 1:

If wetland contains a representative, rare or unique example of a natural or nearnatural wetland type found within the appropriate biogeographic region

Group B

Sites of international importance for conserving biological diversity. *Following "Criteria based on species and ecological communities"*

Criterion 2:

A wetland should be considered internationally important if it supports

• Vulnerable

• Endangered or

• Critically endangered species or

• Threatened ecological communities.

Criterion 3:

A wetland should be considered internationally important if it supports

• Populations of plant and/or animal species important for maintaining the biological diversity of a particular biogeographic region

Criterion 4:

A wetland should be considered internationally important if it supports

o Plant and/or animal species at a critical stage in their life cycles or o Provides refuge during adverse conditions.

Specific criteria based on birds

Criterion 5:

A wetland should be considered internationally important if

• It regularly supports 20,000 birds.

Criterion 6:

A wetland should be considered internationally important if It regularly supports 1% of the individuals in a population of one species or subspecies of water bird.

"Specific criteria based on fish"

Criterion 7:

A wetland should be considered internationally important if

o It supports a significant proportion of indigenous fish subspecies, species or families, life-history stages
o Species interactions.
o And/or Populations that are representative of wetland benefits o And/or Values and thereby contributes to global biological diversity.

Criterion 8:

A wetland should be considered internationally important if

• It is an important source of food for fishes, spawning ground, nursery
• And/or Migration path on which fish stocks, either within the wetland or elsewhere, depend.

"Specific criteria based on other taxa"

Criterion 9:

It regularly supports 1% of the individuals in a population of one species or subspecies of wetland-dependent non-avian species.

36.Management zones of national park ?

Existing use area
• Recreation area
• Cultural/historic area
• Scenic area
• Ecological protected area

37. Write a note on IUCN(5

The International Union for Conservation of Nature (IUCN) is a global environmental organization. It is a neutral forum for governments, NGOs, scientists, business and local communities with the aim of developing solutions and implementing policy related to the environment, conservation and sustainable development.Well-known among its projects is the IUCN Red List of Threatened Species also known as the IUCN Red List or Red Data List. There are various individual Commisions which provide the Union with policy advice on conservation issues.

38.Features of mintaka ? 5

Features

• The terrain is rugged and mountainous
• There is a variety of rocks o Sedimentary o Metamorphic o Igneous Mixture of rock and mineral deposits

Features

• Temperatures at higher elevations remain below freezing point for most of the year

• Winters are severe
• Summers cold and dry.
• Mostly precipitation is in the form of snow.

39.Important of IUCN 2

On the national level, **IUCN helps governments prepare national biodiversity policies**. Internationally, IUCN provides advice to environmental conventions such as the Convention on Biological Diversity, CITES and the Framework Convention on Climate Change. It advises **UNESCO** on natural world heritage.

40.Beauty of Margalla Hill ?

Margalla hills have beautiful torrents gushing down in the monsoon.
• Natural springs are also present.
• The Park includes the Margalla Hills which form's the foothills of the Himalayas, along with Shakarparian Park and Rawal Lake which will increase the beauty of margalla hills.

41.Note on protected area ? 5

Definition:
Wildlife Sanctuary, is a protected area, provides an undisturbed breeding ground for the protection of wildlife. It is a naturally occurring sanctuary, may be an island.
It provides protection for species from
• Hunting
• Predation
• Competition These wildlife refuges are generally officially designated territories. Such refuges can protect animals that are endangered.
Importance:
• National parks, wildlife sanctuaries and other protected spaces, help to conserve the natural world and benefit us in many ways.
• A wildlife sanctuary is promoting conservation, and thus play an important role in society.
Purpose:
Animal sanctuaries exist for many specific reasons:
• Help to protect animals and safeguard their lives.
• Animal sanctuaries give new homes to abandoned animals.
• Animals rescued from dangerous conditions.
• Help to protect animals from illegal activities.
• Serve as places where endangered animals breed and are therefore protected.

42. Area of Lal Suhanra park ?

Lal Sohanra is spread over 162568 acres and is notable for the diversity of its landscape, which includes areas of desert, forest and wetland.

43.What is main objective of national park ?

Primary objective

To protect natural biodiversity along with its underlying ecological structure.

Other objectives

- To manage the area in order to perpetuate, in as natural a state as possible.
- To maintain viable and ecologically functional populations.
- To contribute to local economies through tourism.
- To manage visitor use for inspirational, educational recreational purposes

44.Constitution members ?

Constitution of Board

The Board shall consist of

- Chairman Vice-Chairman
- Number of members as may be appointed by Government.
- Chief Minister of the Punjab shall be the chief warden and *ex-officio* Chairman of the Board.
- Minister incharge of the Department shall be the *ex-officio* Vice Chairman of the Board.
- Secretary to Government, Forestry, Wildlife, Fisheries and Tourism Department shall be the *ex-officio* Secretary of the Board.

45. Location of Astore ?

Location:

- Astore District
- Between Nanga Parbat
- Gilgit–Baltistan, Pakistan

46.Major Challenges of veterinarians ?

Major Challenges to Veterinarians

- Environmental contaminants
- Infectious diseases
- Outbreaks that potentially threaten wild and domesticated animals as well as humans.

Wildlife veterinarians are necessary and should made available to care of wild animals.

47. Write the name of any 4 national parks ?

1 : Ayub Park Rawalpindi
2. Shandure-Phander National Park Ghizer
3. Broghil Valley Chitral
4. Central Karakoram Gilgit and Skardu

48.Wild plant products ?

variety of plant products like lumber, paper, rubber, oils, dyes, and fibers etc. are supplied by wild plants.

49.Any two experimental wild animals ?

1 : ntelope

2 : Monkey

50.Name of mammals live in margalla national park ?

- Indian leopard.
- Jungle cat.
- Leopard cat.
- Central Asian boar.
- Golden jackal.
- White-footed fox.
- Rhesus macaque.

51.What are issues to tunsa and chashma barrage dolphins sanctuary ?

Issues for consideration:

• Taunsa Barrage Wildlife Sanctuary is currently confronted with multiple issues
• Limited cultivation is practiced inside the sanctuary area
• Clearing off the natural vegetation
• Alteration of the natural ecosystem
• Degrading wildlife habitat
• Invasion of alien species
• Hence, adverse environmental impacts

52.Name of polo grounds of tooshi long ?

Tooshi, a conservation area for Markhor. It is famous for the polo grounds of

- Shandur
- Broghil National Park
- Mastuj
- Booni

Bamborat: area where the Kalash cultural festival is held every August of the year.

53.Wetland significant ? L-5

Wildlife is an integral part of an ecosystem.

- Human himself as animal inter-related and inter-dependent on wildlife for its survival.
- From clothes to medicines, building materials to variety of chemicals all are extracted from exiting wildlife around us.
- About 90% of the domestic food crops now cultivated are descendants of wild plants.
- Today, variety of plant products like lumber, paper, rubber, oils, dyes, and fibers etc. are supplied by wild plants.
- Plants recycle nutrients necessary for agricultural production and help maintain soil fertility.
- Wild plants or their extracts are a source of medicine for an estimated 80% of the world's population. ➢ Source of oxygen.

Insects and birds add to the economic gains through pollination of plants.

- Maintaining the 'balance of Nature'. e.g. Carnivore-herbivore balance.

54.Wetlands define ?

Wetlands are defined according to Ramsar Convention as

"Areas of marsh, fen, peat land or water, whether natural or artificial, permanent or temporary,with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six meters".

Define 2 : Wetlands are areas where water is the primary factor controlling the environment and the associated plant and animal life.

55.Difference endangered and extinct ?

Endangered species	Extinct species
The species which are at the verge of extinction and which are required to be protected and conserved are known as endangered species.	**The species which no more exist on the earth are known as extinct species.**
Example: tiger, wild buffalo etc.	**Example: dinosaurs**

56. Constitution board (3)

Constitution of Board

The Board shall consist of

• Chairman

• Vice-Chairman

• Number of members as may be appointed by Government.

• Chief Minister of the Punjab shall be the chief warden and *ex-officio* Chairman of the Board.

• Secretary to Government, Forestry, Wildlife, Fisheries and Tourism Department shall be the *ex-officio* Secretary of the Board.

57. Ayubia major bird fauna (3)

Major Birds Fauna:

• The park harbors up to 203 species of birds.

• Golden eagle

• Vulture

• Eurasian sparrow

• Hawk

• Hill pigeon

• Khalij Pheasant (30 species)

• Spotted dove

• Collared dove

58.Small Grants Programme SGP (5)

Small Grants Programme (SGP)

• *"Small support for long term impact".*

SGP of WWF-Pakistan has been supporting grass root level organizations and

researchers to strengthen efforts to conserve threatened and endangered species and ecosystem. Under the programme more than 450 projects have been successfully completed.

SGP has supported innovative conservation initiatives for priority species and ecosystem.

- Marine turtles
- Snow leopard
- Indus River dolphin
- Sharks
- Mangroves forests
- Indus river delta Collaborative research programs should be initiated with neighboring countries with common ecosystem.

59.What is SSC ? 2

The **SSC** is one of the commission of IUCN, the World Conservation Union.

Mandate of the Commission :

The Commission shall conduct its activities in accordance with its mandate as established by the World Conservation Congress.

60.What is markhor ? 2

The markhor is the National Animal of Pakistan , where it is also known as the screw horn or " screw-horned goat" The word "Markhor" is Persian for snack eater. They mostly live in the Northern Areas of Pakistan especially in the Chitral , Ghilzar and Hunza regions. There are about 2000-4000 Markhor in the wild.

61.Area of hub dam?

27,219ha

62.Role of animal hygein environment?

63. Location of Margalla hills ?

Location:

North of the Islamabad City, Capital of Pakistan.

64. National park 3 names..

1. Ayub Park Rawalpindi
2. Shandure-Phander National Park Ghizer
3. Broghil Valley Chitral

65. pollution factors.

Pollution of various kinds have adversely affected wild animals.

- **Insecticides**

There has been a considerable damage to birds by insecticides spraying.

- **Oil Spillage**

Oceanic birds are killed by oil spillage from oil tankers in the sea.

- **Noise Pollution**

It causes disturbance of hormonal and reproductive behavior of many wild animals.

- **Water Pollution**

Domestic as well as industrial waste dumped into water bodies causes decline in stocks of fishes.

- Animals like mussels have been unfit for human consumption

- Various natural calamities like floods, hurricane, fire etc. have also caused massive destruction of wildlife.

67. Primary objective to protected areas.

- To protect natural ecosystems
- Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

68. History of Ramsar

It is one of the global inter-governmental environmental agreements. The treaty was negotiated in1960s by countries and N. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In a 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975,

69. Concept of Wildlife Sanctuaries

Definition:

Wildlife Sanctuary, is a protected area, provides an undisturbed breeding ground for the
protection of wildlife. It is a naturally occurring sanctuary, may be an island.

It provides protection for species from

- Hunting
- Predation
- Competition

These wildlife refuges are generally officially designated territories. Such refuges can protect animals that are endangered.

Importance:

- National parks, wildlife sanctuaries and other protected spaces, help to conserve the natural world and benefit us in many ways.
- A wildlife sanctuary is promoting conservation, and thus play an important role in society.

Purpose:

Animal sanctuaries exist for many specific reasons:

- Help to protect animals and safeguard their lives.
- Animal sanctuaries give new homes to abandoned animals.
- Animals rescued from dangerous conditions.
- Help to protect animals from illegal activities.
- Serve as places where endangered animals breed and are therefore protected.

70. Define Game Reserve ? 2

An area where regulated hunting & shooting is allowed.

71. Gene Bank ? 2

Gene banks are the bio-repositories used for the preservation of genetic material.

❖ Wildlife can be served as a source of Gene Bank.

72. Area of lal suhanra national park. 2

Lal Suhanra is spread over 31,368 hectares Notable for the diversity of its landscape, which includes areas of desert, forest and wetland.

73. what is the concept of natural park ? 2

- National park, an area set aside by a national government for the preservation of the natural environment. A national park may be set aside for purposes of public recreation and enjoyment or because of its historical or scientific interest.

74. Where the cholistan sanctuary is located.

Cholistan Wildlife Sanctuary is part of the Cholistan desert.

Location: Cholistan desert. In the South Eastern portion of Punjab.

75. Wildlife Census

For implementation of various protective laws an accurate idea about population density of species is essential.
• Censuses require a lot of work, but these efforts serve an important purpose.
Census is mandatory to count numbers and distribution of wildlife populations.
• Proper planning
• Implementation
• Evaluation of conservation programs
• African elephant, which are under escalated threat from poaching, habitat loss and human–wildlife conflict.
• An aerial census conducted in the border between Kenya and Tanzania.
Census can be helpful in mapping factors of
• Animals roaming
• Water sources
• Human activities, such as livestock grazing and charcoal burning. [07:52, 15/01/2022]

76. IWC stands for ?

(international whaling commission)

77. name Endangered species in Pakistan ?

- Markhor.
- Mountain Weasel.
- Asian Black Bear. people
- Baluchistan Forest Dormouse
- Black Finless Porpoise.
- European Otter
- Vole
- Indus River Dolphin.

- Caracal
- Bigeye Tuna